

Key GitHub Repos & Papers - Quick Reference

■ MUST FORK (Start This Week)

1. Mip-Splatting (Anti-Aliasing)



Quick Start:



2. GaussianEditor (Scene Editing)



Quick Start:



3. Original 3D Gaussian Splatting



■ HIGH PRIORITY (Months 2-4)

4. 4D Gaussian Splatting (Animation)



Anime Use Case: - Flowing hair animation - Moving clouds/water - Character idle animations

5. GaussianAvatars (Characters)



Anime Adaptation: - Train on VRoid character models - Adapt for stylized proportions - Add anime-specific expressions

6. SceneDreamer (Unbounded Worlds)





Anime Use Case: - Extend single anime background infinitely - Generate Ghibli-style landscapes - Procedural world generation

■ IMPORTANT (Months 4-6)

7. DreamGaussian (Generation)



Quick Integration: - Already has Gradio demo - Export to PLY for your viewer - Fine-tune on anime assets

8. Nerfstudio (Framework)



Why Use: - Modular architecture - Real-time web viewer - Easy to prototype new ideas - Supports 3DGS (Splatfacto method)

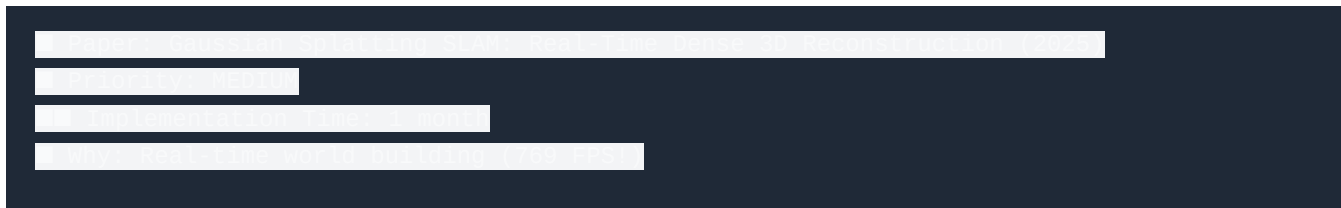
9. gsplat (Efficient 3DGS Library)



Performance: - 15% faster training - 4x less GPU memory - Python bindings

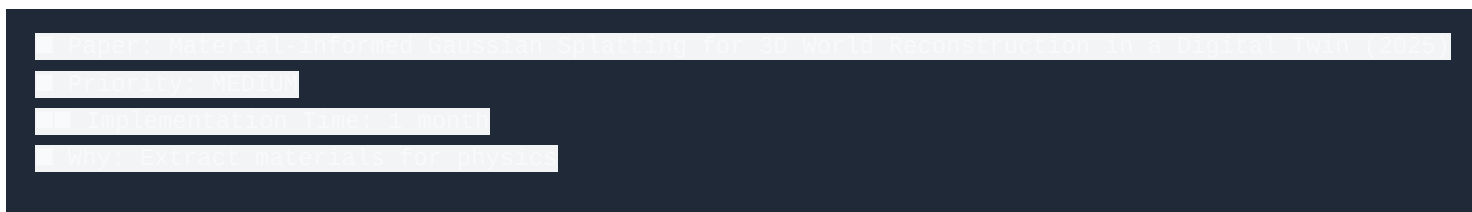
■ Key Papers (No Code Yet - Need R&D)

10. Gaussian Splatting SLAM



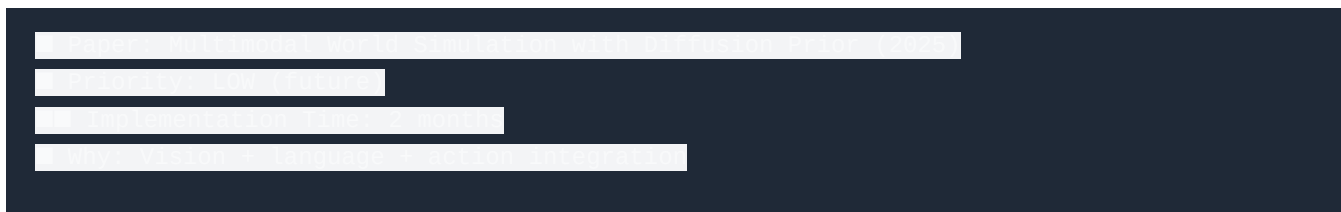
Status: Paper available, implementation unclear **Action:** Contact authors or implement from paper

11. Material-Informed Gaussian Splatting



Status: Submitted to IEEE IV 2026 **Action:** Wait for code release or implement from paper

12. Multimodal World Simulation with Diffusion Prior

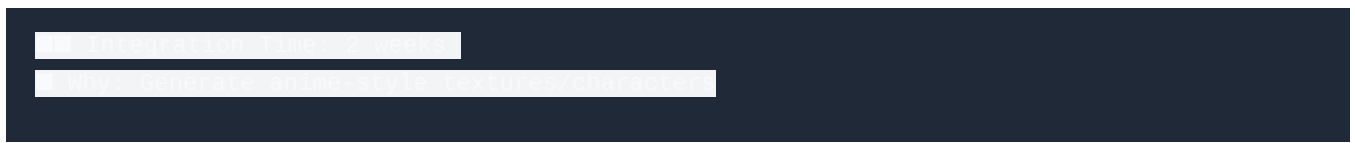


Status: arXiv preprint **Action:** Monitor for code release

■ Anime-Specific Resources

Stable Diffusion (Anime Fine-tuning)





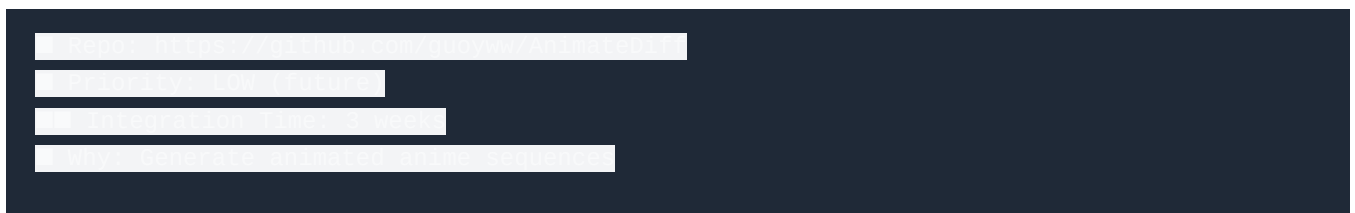
Key Models: - Anything v5 (anime style) - Counterfeit v3 (clean anime) - AbyssOrangeMix (vibrant colors)

ControlNet (Guided Generation)



Anime Use Cases: - Character pose control - Maintain anime line art - Scene composition

AnimateDiff (Video Generation)



Integration: - Use for training 4DGS - Generate character animations - Create ambient scene motion

■ ■ Supporting Tools

COLMAP (Structure from Motion)



Zero123 (Multi-view Generation)



rembg (Background Removal)



SAM2 (Segmentation)



■ Comparison Matrix

Technology	Readiness	Integration Time	Anime Suitability	Impact
Mip-Splatting	9/10	1 week	■■■■■	Quality
GaussianEditor	9/10	2 weeks	■■■■■	Interaction
GaussianAvatars	8/10	1 month	■■■■	Characters
4DGaussians	8/10	1 month	■■■■■	Dynamics
SceneDreamer	8/10	6 weeks	■■■■	Scale
DreamGaussian	8/10	2 weeks	■■■	Generation
Nerfstudio	8/10	2 weeks	■■■	Framework
GS-SLAM	7/10	1 month	■■■■	Real-time
Material-GS	7/10	1 month	■■■	Physics
World Models	6/10	2 months	■■■■	Intelligence

■ Integration Order (Recommended)

Month 1:

1. **Mip-Splatting** → Quality improvement
2. **GaussianEditor** → Basic interaction

Month 2:

1. **Stable Diffusion + LoRAs** → Anime generation
2. **Zero123** → Multi-view synthesis

Month 3:

1. **4DGaussians** → Animation prototype
2. **ControlNet** → Generation control

Month 4:

1. **GaussianAvatars** → First character
2. **SceneDreamer** → Unbounded worlds

Month 5:

1. **DreamGaussian** → Fast generation
2. **SAM2** → Segmentation for editing

Month 6:

1. **GS-SLAM** → Real-time reconstruction
2. **Material-GS** → Physics integration

■ Community Resources

Discord Servers:

- **Nerfstudio**: <https://discord.gg/nerfstudio>

- **3DGS Community:** Check graphdeco-inria/gaussian-splatting issues
- **Stable Diffusion:** <https://discord.gg/stablediffusion>

Reddit:

- r/MachineLearning (technical)
- r/StableDiffusion (generation)
- r/anime (user feedback)
- r/virtualreality (VR integration)

Twitter Accounts to Follow:

- @_akhaliq (daily paper summaries)
- @hardmaru (world models, RL)
- @poolio (graphics, rendering)
- @DrJimFan (embodied AI)

YouTube Channels:

- Károly Zsolnai-Fehér (Two Minute Papers)
- Yannic Kilcher (paper explanations)
- AI Coffee Break (accessible ML)

■ Red Flags to Watch For

Technical:

■ **Repo abandoned** (no commits in 6+ months) ■ **No license** (legal issues for commercial use) ■ **Only paper, no code** (need to implement from scratch) ■ **Requires rare hardware** (TPUs, exotic GPUs)

Performance:

■ **< 30 FPS on RTX 3080** (too slow for real-time) ■ **> 5 min generation time** (poor UX) ■ **> 24GB VRAM required** (limits scalability)

Quality:

■ **Anime style lost in conversion** (need custom training) ■ **Inconsistent multi-view** (breaks immersion) ■ **Visible artifacts** (seams, holes, flickering)

■ Pro Tips

When Forking:

1. **Check "Watch releases"** → Get notified of updates
2. **Star the repo** → Track for future reference
3. **Read all Issues** → Common problems + solutions
4. **Check Pull Requests** → Community improvements

When Training:

1. **Start small** → 100 images, not 100K
2. **Visualize early** → Don't wait for full training
3. **Save checkpoints** → Every 1K iterations
4. **Log everything** → Use Weights & Biases or TensorBoard

When Debugging:

1. **Isolate** → Test each component separately
 2. **Simplify** → Remove features until it works
 3. **Compare** → Run both old and new versions
 4. **Ask** → Post issues on GitHub, don't suffer alone
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■ Learning Resources

Beginner (3DGS Fundamentals):

1. **Two Minute Papers**: "3D Gaussian Splatting Explained"
2. **Károly's Blog**: Technical deep dive
3. **Original 3DGS paper**: Read sections 1-3

Intermediate (Implementation):

1. **Nerfstudio tutorials:** Hands-on training
2. **3DGS Discord:** Ask questions
3. **GitHub Issues:** See how others solved problems

Advanced (Research):

1. **CVPR 2024 papers:** Latest methods
2. **ArXiv daily:** New preprints
3. **SIGGRAPH Asia 2024:** Upcoming work

■ Starter Pack (Clone These Today)



Total Download Size: ~5-10 GB **Setup Time:** 2-3 hours **Value:** Priceless ■

■ Success Metrics

After integrating each repo, measure:

1. **Performance:** FPS at 1080p
2. **Quality:** PSNR, user rating
3. **Speed:** Generation time
4. **Memory:** GPU VRAM usage
5. **Stability:** Crash rate

Target: All metrics green before moving to next repo

■ Final Checklist

This Week: - ☐ Star all repos above - ☐ Clone Mip-Splatting + GaussianEditor - ☐ Read their READMEs thoroughly - ☐ Join Nerfstudio Discord

This Month: - ☐ Integrate Mip-Splatting - ☐ Setup GaussianEditor demo - ☐ Train on 100 anime images - ☐ Share demo on Twitter

This Quarter: - ☐ Add 4DGS animation - ☐ Create first rigged character - ☐ Launch public beta - ☐ Get 1000 users

Your anime world awaits. Clone a repo TODAY. ■■

Last Updated: 2026-01-07 Check back weekly for new papers/repos